

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0376

Ninemile Point 138 kV
September 26, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0376. The Interconnection Customer proposes a 229.0 MW solar photovoltaic facility interconnecting at the Ninemile Point 138 kV substation in the Delaware area. The request was received on September 05, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 238 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,289,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: NINEMILE PT to NLEBANON 138.0 kV
Contingency: N LEBANON to NINEMILE POINT 138.0 kV
Loading: 185.4% (Rating: 288 MVA)
DFAX: 62.7%
Pre-project loading: 135.5%
2. Facility: NINEMILE PT to N LEBANON 138.0 kV Ckt 1
Contingency: NORTH LEBANON SUBSTATION to NINEMILE PT 138.0 kV #2
Loading: 181.5% (Rating: 294 MVA)
DFAX: 62.7%
Pre-project loading: 132.6%
3. Facility: SENECA to MCKEE RUN 138.0 kV
Contingency: VERNON to LIMON 138.0 kV Line
Loading: 172.0% (Rating: 209 MVA)
DFAX: 5.0%
Pre-project loading: 166.5%
4. Facility: FAIRONT to N LEBANON 138.0 kV Cir. 2
Contingency: NORTH LEBANON to FAIRMONT 138.0 kV Ckt 1
Loading: 158.5% (Rating: 316 MVA)
DFAX: 61.7%
Pre-project loading: 113.8%
5. Facility: N LEBANON to FAIRMONT 138.0 kV Line 1
Contingency: NORTH LEBANON to FAIRMONT 138.0 kV Line 2
Loading: 156.6% (Rating: 320 MVA)
DFAX: 61.7%
Pre-project loading: 112.4%

6. Facility: BOND-ELK GROVE FLORIN to FAIRMONT 138.0 kV Circuit 1
Contingency: BOND-ELK GROVE FLORIN to FAIRMONT 138.0 kV Line 2
Loading: 153.9% (Rating: 322 MVA)
DFAX: 62.3%
Pre-project loading: 109.5%
7. Facility: BOND-ELK GROVE FLORIN to FAIRMONT 138.0 kV
Contingency: BOND-ELK GROVE FLORIN to FAIRMONT 138.0 kV Circuit 1
Loading: 145.8% (Rating: 339 MVA)
DFAX: 62.3%
Pre-project loading: 103.7%
8. Facility: CHURCHS FERRY to WILLOW PASS 138.0 kV
Contingency: DELAWARE CITY PLANT to RED LION ENERGY CENTER SUBSTATION 138.0 kV L
Loading: 135.1% (Rating: 196 MVA)
DFAX: 8.6%
Pre-project loading: 125.1%
9. Facility: DELL CRK to CHURCHS FERRY 138.0 kV
Contingency: RED LION ENERGY CENTER to DELAWARE CITY PLANT 138.0 kV
Loading: 121.5% (Rating: 142 MVA)
DFAX: 6.4%
Pre-project loading: 111.1%
10. Facility: WHITMORE to BOND-ELK GROVE FLORIN 138.0 kV line
Contingency: BOND-ELK GROVE FLORIN 345-138 kV TX
Loading: 113.3% (Rating: 274 MVA)
DFAX: 29.3%
Pre-project loading: 88.8%
11. Facility: WILLOW PASS to BANTA CARBONA 138.0 kV
Contingency: BOND-ELK GROVE FLORIN 345 kV/138 kV Autotransformer
Loading: 110.5% (Rating: 259 MVA)
DFAX: 18.2%
Pre-project loading: 94.4%
12. Facility: SENECA to BANTA CARBONA 138.0 kV Line
Contingency: LIMON to VERNON 138.0 kV Line
Loading: 105.2% (Rating: 277 MVA)
DFAX: 5.0%
Pre-project loading: 101.0%
13. Facility: EDGEWATER STA to WHITMORE 138.0 kV L
Contingency: BOND-ELK GROVE FLORIN 345-138 kV Transformer
Loading: 103.5% (Rating: 300 MVA)
DFAX: 29.3%
Pre-project loading: 81.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NINEMILE PT to NLEBANON 138.0 kV	Rebuild line with higher-capacity conductor	\$2,011,000	38% (\$768,000)
NINEMIL POINT to NORTH LEBANON 138.0 kV	Rebuild line with higher-capacity conductor	\$2,297,000	38% (\$877,000)
MCKEE RUN to SENECA 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,515,000	36% (\$2,009,000)
FAIRONT to N LEBANON 138.0 kV Cir. 2	Rebuild line with higher-capacity conductor	\$4,790,000	38% (\$1,828,000)
NORTH LEBANON to FAIRMONT 138.0 kV Circuit 1	Rebuild line with higher-capacity conductor	\$4,714,000	38% (\$1,799,000)
BOND-ELK GROVE FLORIN to FAIRMONT 138.0 kV	Rebuild line with higher-capacity conductor	\$2,059,000	38% (\$786,000)
BOND-ELK GROVE FLORIN to FAIRMONT 138.0 kV	Rebuild line with higher-capacity conductor	\$2,933,000	38% (\$1,119,000)
CHURCHS FERRY to WILLOW PASS 138.0 kV	Rebuild line with higher-capacity conductor	\$1,723,000	41% (\$707,000)
DELL CRK to CHURCHS FERRY 138.0 kV	Rebuild line with higher-capacity conductor	\$2,633,000	46% (\$1,222,000)
WHITMORE to BOND-ELK GROVE FLORIN 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,725,000	34% (\$1,925,000)
WILLOW PASS to BANTA CARBONA 138.0 kV	Rebuild line with higher-capacity conductor	\$23,954,000	31% (\$7,327,000)
SENECA to BANTA CARBONA 138.0 kV Line	Reconductor line	\$1,159,000	36% (\$422,000)
EDGEWATER STA to WHITMORE 138.0 kV L	Reconductor line	\$1,487,000	34% (\$500,000)
	Total allocated to project		\$21,289,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.